The design of `GaugeTheory/`

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1 Purpose

`PhyslibAlpha/ClassicalFieldTheory/GaugeTheory/` is the symbolic framework for the local content of a gauge theory: the algebra in which a Lagrangian is written, the gauge and Lorentz actions on it, and the invariants of a given mass dimension. Fields enter only through their jets at one point, the formal symbols $\partial_s \psi_\alpha$.

One rule organises the folder: **every layer is parametrised by data and returns constructions and theorems; a concrete theory contributes only its data.** A model supplies a gauge datum and a table of matter fields, and everything else about it is inherited.

The target theorem. For the Standard Model, the gauge- and Lorentz-invariant elements of the local field algebra of mass weight at most 8 (mass dimension at most 4) are spanned by an explicit list. This is a classification of invariant *polynomials*: it is over $\mathbb{C}$, imposes no reality condition, and takes no quotient by total derivatives or equations of motion. It therefore contains terms such as $\partial^2(H^\dagger H)$ and $\operatorname{tr} F\tilde{F}$, and it should be named as a statement about invariants, not as "the Lagrangian". Reality and the quotient by total derivatives are later steps (§6).

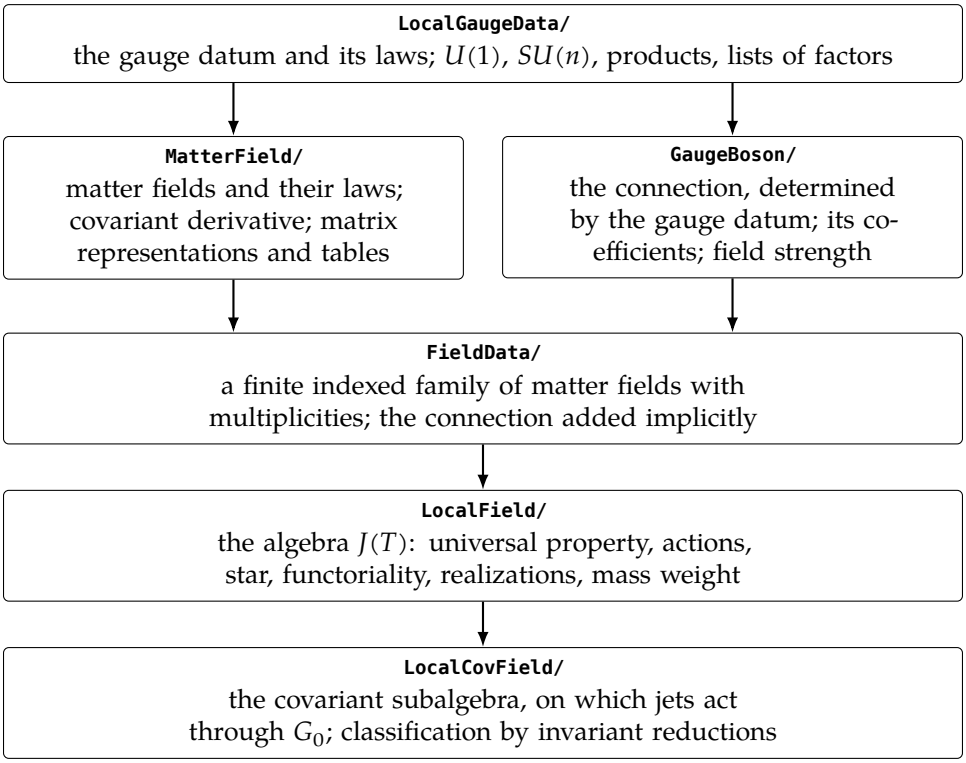
Deadline. Early December 2026, with the end of the semester as the hard deadline.

2 The idea

- **Gauge jets.** A gauge transformation is its jet U in a jet group G_J: value $\operatorname{eval} U \in G_0$, Lie algebra $\mathfrak{g}_J$, derivatives, adjoint action, Maurer–Cartan form $\omega_\mu(U) = i(\partial_\mu U)U^{-1}$. These maps and their laws are the gauge datum; nothing downstream knows which group it is.
- **Matter fields.** A matter field is a finite-dimensional complex space V with a Lorentz representation, an action of G_J on $\texttt{JetRing} \otimes V$ that is fibrewise and generated by an infinitesimal action of $\mathfrak{g}$, a mass weight and a statistic (bosonic or fermionic). Its symbols transform linearly.
- **The gauge field.** The connection is not listed by a theory: it is determined by the gauge datum. Its symbols transform affinely, by the adjoint action plus the Maurer–Cartan form, and the translation part satisfies the cocycle law $t(UV) = t(U) + \operatorname{Ad}_U t(V)$. Matter and the connection are two separate notions; the affine picture describes how they differ but is not a common type.
- **Field data.** A theory's field data is a finite indexed family of matter fields, each with a multiplicity (its number of generations). The connection is added from the gauge datum, so the pure gauge theory is the field data with no matter, and the sum of two theories is the disjoint union of their matter families.

- **The algebra.** $J(T)$ is the free graded-commutative algebra on the symbols of the matter fields of T, their conjugates, and the connection. Every action is defined on generators and extended by the universal property; the translation part of the connection gives the inhomogeneous term. A map of index types gives an equivariant map of algebras, and sectors are its images.
- **Invariants.** Mass weight is twice the mass dimension (derivative 2, scalar 2, Weyl fermion 3, connection 2). On the covariant subalgebra jets act through G_0, so classification is ordinary invariant theory. It is done one family of terms at a time: a *reduction* ReducesInvariantsTo $\sigma\ V\ W$ says that for every σ-stable submodule S, an invariant of $V \sqcup S$ lies in $W \sqcup S$. Reductions compose, combine over finite joins and transport along maps of index sets, and σ is any family of linear maps, so gauge and Lorentz invariance are one invariance for the disjoint union of the two groups.

3 The layers



Imports go down the line only. Three points decide the shape:

- **Two field notions, one interface per model.** Matter fields and the connection are separate: each states its own transformation laws once, and the matter laws are stated for an arbitrary `MatterField`. A statement over all fields of a theory uses a type built once the model is fixed,

```
inductive Field (T : FieldData jets) | matter (i : T.Idx) | gauge
```

  which lives in `Type`, has decidable equality, and whose laws dispatch to the matter and connection laws. A closed type `Species | matter M | gauge` at the level of data was tried (PR #2011) and dropped: every general lemma split into the two cases, the matter case only restated existing results, and it did not say how a theory contains its gauge field.

- **One algebra tower.** The pure gauge theory is J of the field data with no matter, and the sectors of $J(T)$ are images of sub-families. Nothing is built twice and identified afterwards.

- **One classification mechanism.** The relation `ReducesInvariantsTo` of `Mathematics/InvariantReduction` is general mathematics and stays there. The algebra layer supplies what every theory needs to use it: the gauge and Lorentz actions as one family indexed by $G_0 \sqcup SL(2,\mathbb{C})$, the reduction of a factor's invariance to that family, and the stability of symbol ranges. A classification theorem for one index law (a colour trace, an isospin contraction, a Lorentz scalar) is stated as a reduction modulo an arbitrary stable submodule, so it can be applied inside any larger expression and to any extension.

Each layer is described below by its datum, what it derives and its interface.

3.1 `LocalGaugeData/`: the gauge datum

The question it answers. A Lagrangian density at a point x depends on the fields and finitely many of their derivatives at x. So when we ask whether it is gauge invariant, a gauge transformation $g(x)$ matters only through its Taylor series at that point, its *jet*. This layer describes these jets and nothing else. It does not mention fields or Lagrangians; every later layer takes its datum as input.

An example first. For $U(1)$, a gauge transformation is $u(x) = e^{i\theta(x)}$. Its jet is the power series of u in the coordinates $x_0, \dots, x_3$, a unitary element of $\mathbb{C}[[x]]$. Its value at the base point is $u(0) \in U(1)$. Its *Maurer–Cartan form* $\omega_\mu = i(\partial_\mu u)u^{-1} = -\partial_\mu\theta$ is exactly the shift of the gauge field, $A_\mu \mapsto A_\mu + \omega_\mu$. For $SU(n)$ everything is the same with $n \times n$ matrices of power series, and $\omega_\mu = i(\partial_\mu U)U^{-1}$ is a traceless hermitian matrix.

The datum. `LocalGaugeData` G_0 $\mathfrak{g}$ `GJ` $\mathfrak{g}_J$ abstracts the example. It has four types: the gauge group G_0 and its Lie algebra $\mathfrak{g}$, and their jet versions G_J and $\mathfrak{g}_J$. It has the maps a transformation law uses:

- evaluation at the base point and constant jets, $G_J \leftrightarrows G_0$ and $\mathfrak{g}_J \leftrightarrows \mathfrak{g}$;
- derivatives ∂_μ and multiplication by coordinates x_μ on $\mathfrak{g}_J$;
- the adjoint action Ad_U of a gauge jet on Lie algebra jets;
- the Maurer–Cartan form $\omega_\mu(U)$.

It also has the rules these satisfy in the example, such as $\omega(UV) = \omega(U) + \mathrm{Ad}_U\,\omega(V)$ and $\partial_\mu\omega_\nu - \partial_\nu\omega_\mu + [\omega_\mu, \omega_\nu] = 0$. Everything later is proved from these rules alone, so it holds for every gauge group at once.

What is proved from the rules. Three tools that later layers need:

- how Ad_U acts on Taylor coefficients (a Leibniz formula), which is how a gauge jet moves the derivatives of a field;
- the *symmetrised* Maurer–Cartan data, a smaller set of numbers from which all Taylor coefficients of ω can be recovered;
- the jets that agree with the identity up to order n; these form normal subgroups, and such a jet only shifts the first n derivatives of the gauge field.

Two extra properties. The rules do not say that G_J really consists of Taylor series, so two properties are stated separately. `Faithful`: a jet is determined by its Taylor coefficients. `Free`: every choice of Taylor coefficients comes from some jet. `Free` is what later shows that the gauge field enters an invariant only through its field strength: gauge jets are rich enough to remove every other combination of derivatives of A_μ.

Where concrete groups come from. A *matrix presentation* (`MatrixJets`) says that the four types sit inside matrices and every map is the matrix one. From it the rules and `Faithful` are proved once, and `Free` is reduced to three conditions on the carriers, so $SU(n)$ (`su n`) and $U(1)$ (`u1`, as 1×1 matrices) are obtained by filling in a presentation. Products (`prod`) and a list of factors (`ofFactors`) then give any group such as $SU(3) \times SU(2) \times U(1)$, written `[.SU 3, .SU 2, .U1]`.

Factors and morphisms. Charges are given one factor at a time, for example "a doublet of $SU(2)$ with hypercharge -3". A *factor* (`U1Factor`, `SUFactor`) records how the gauge jets project onto one $U(1)$ or $SU(n)$. A *morphism* (`Hom`) is a map between two data that respects all the maps above, and a factor of the target pulls back along it. The projections of a product are morphisms, which is how the factors of $SU(3) \times SU(2) \times U(1)$ are obtained from those of its pieces.

Conventions. The datum is always passed explicitly, never found by instance search, because the four types do not determine it. The layer imports nothing from the layers above it.

3.2 `MatterField/` and `GaugeBoson/`: the two kinds of field

Matter. `MatterField jets` has a complex value space V (free and finite), a Lorentz representation, the jet action `repJet` on `JetRing` $\otimes V$, required to be fibrewise and generated by the infinitesimal action `repAlgebra`, a mass weight and a statistic (PR #1989). Matrix representations and tables (`MatrixRep`) produce matter fields from charges one factor at a time and prove the infinitesimal-action condition once for all of them. The coefficients by which a jet moves the derivatives of a field live in `RepCoeff`, separate from the covariant derivative, so that field data can use them without importing the algebra.

The connection. The value space is CoVector $\otimes\, \mathfrak{g}$, real. The coefficients are the adjoint transport and the Maurer–Cartan coefficients (`GaugeCoeff`), with their composition laws: the linear part composes by a Leibniz sum, and the translation part is a cocycle.

Requirements from the classification. Keep the value-space representation distinct from the dual action on field symbols, including their conjugates. Parity and mass weight belong to the field; bases and particular fundamental, conjugate or adjoint representations are inputs to specific classifiers, not requirements on every field. Keep the conditions behind covariance visible: `GaugeLorentzCompatible` supplies Lorentz stability of covariant expressions, and `PureJetsActTrivially` makes their jet action factor through evaluation. Whether these become fields of `MatterField`, discharged once for matrix representations, or remain hypotheses on the field data is open (§5). The converse, that jet-invariant expressions lie in the covariant algebra, uses `Free` and is a separate result.

3.3 `FieldData/`: what a theory contains

The field data is the indexed family that exists today as `GaugeFieldData`: index types `FermionSpecies` and `BosonSpecies` with `Fintype` and `DecidableEq`, a `MatterField` for each index with the matching statistic, and the gauge field implicit. It may be renamed `FieldData`. The model card's table, which lists named fields with their generation counts, compiles into it.

Two alternatives were considered and rejected as the definition:

- **One bosonic and one fermionic field**, merged by `MatterField.prod` or `pi`. Merging forces one mass weight per statistic, forgets the names and generations (two copies of one multiplet, as in a two-Higgs-doublet model, cannot be told apart), has no empty case, and no lemma yet

shows the compatibility conditions survive it. The merged fields remain available as a derived view, with the common-weight hypothesis supplied where they are used.

- **A multiset of matter fields.** `MatterField` contains a type and has no decidable equality, so counting and `decide` fail, and identical multiplets collapse into one entry.

Fields with an affine shift other than the connection (Stückelberg, axion), if ever needed, get their own family with an affine datum.

3.4 `LocalField/` and `LocalCovField/`: the algebras

$J(T)$ is built from the per-field generator spaces, bosonic generators commuting and fermionic ones anticommuting, with the connection's generators complexified. Its universal property says that every compatible assignment of the symbols to an associative complex algebra B extends to exactly one algebra map; B need not be commutative and the map need not be injective. Gauge and Lorentz actions, ordinary derivatives and mass weight are defined on generators. The current sectors use only the totals per category (fermion, gauge, scalar); sectors by sub-families follow from maps of index types.

The covariant subalgebra is generated by the matter fields, their symmetrised covariant derivatives and the covariant derivatives of the field strength; on it the jets act through G_0. Each sector of the classification supplies a reduction, containment of its classified span in the source, and pointwise fixedness of that span; these combine without sector independence.

Conjugates and reality. The conjugate components already sit in each field's jet component space. Still to decide (§5): a real/complex flag per matter field, conjugate symbols generated and never listed, and a star structure on $J(T)$ that swaps a symbol with its conjugate and fixes real fields and the connection. A real Lagrangian is then a star-fixed invariant.

4 What a model sees

A model is a card and nothing else. The Standard Model card (`PhyslibAlpha/Particles/StandardModel/Basic.lean`) reads:

```
abbrev gauge : List FactorSpec := [.SU 3, .SU 2, .U1]
abbrev leptonDoublet : MatterFieldData factors := (.L, .singlet, .fund, -3)
abbrev higgs : MatterFieldData factors := (.scalar, .singlet, .fund, 3)
inductive Fields | q | l | u | d | e | H
def table : FieldData factors Fields | .q => (3, quarkDoublet) | ...
```

The fields of the model are abbreviations of the value spaces the card produces, and theorems are stated in the card and the generic vocabulary alone. This is the interface of model-building tools such as SARAH and Sym2Int: the gauge group and a list of matter fields with generations, the gauge fields never listed. To complete it:

1. real fields: a real-scalar label or a reality flag (`LorentzLabel` has only `L`, `R`, `scalar`, all complex);
2. a gauge field and a field strength for each gauge factor, obtained from the factor projections, with names the card can use;
3. conjugates generated automatically, through the star structure and a user-facing conjugation;
4. writing a term from the card's symbols and checking its invariance (the realization of symbols in $J(T)$);

5. symmetry breaking (vacuum expectation values, mass eigenstates; pulling back along a `Hom` already gives the branching of each field) and global or discrete symmetries, including the flavour group acting on generations.

Items 1–3 belong with the field-data work; item 4 is the realization PR; item 5 comes after the deadline.

5 Decisions and open questions

question	answer	status
One notion for matter and the connection, or two?	Two. A common interface exists only once the model is fixed (`Field T`).	agreed, 2 Nov
Connection explicit in the field data or implicit?	Implicit, from the gauge datum. Sums are disjoint unions of matter families.	agreed, 2 Nov
Shape of the field data	The indexed families of `GaugeFieldData`; merged fields only as a derived view.	proposed
What happens to PR #2011	Keep the coefficient extraction; move the gauge-field composition laws and the component form of the gauge law to `GaugeBoson/GaugeCoeff`; drop the species type.	proposed
Real or complex tower	One complex tower with a star, real objects being its star-fixed parts; the real connection algebra stays until step 4 of §6.	proposed
Sectors	By maps of index types. A grading by sub-families waits until a proof needs it.	proposed
Conjugates and reality	Reality flag, generated conjugates and a star, as in §3.4.	open
Compatibility conditions	Fields of `MatterField` or hypotheses on the field data.	open
Names	`MatterField`, `FieldData` (renaming `GaugeFieldData`), `Field T`.	proposed

The two open questions should be settled before the realization of symbols in $J(T)$.

6 Distance from the current tree

current	intended
`LocalGaugeData/`	the same (done)
`MatterField/`, `GaugeBoson/Basic`, `GaugeBoson/GaugeCoeff`	the same; the gauge-field results of PR #2011 join `GaugeCoeff`
`GaugeBoson/LocalGauge/`, `GaugeBoson/LocalGaugeCov/` (the real connection tower, about 3300 lines)	the two algebra layers at the field data with no matter
`GaugeFieldData/`	`FieldData/`, the same structure, with `Field T`
`LocalField/`, `LocalCovField/`	the same, sectors derived from maps of index types
`Mathematics/InvariantReduction`; `StandardModel/InvariantReduction`	the same; the combined family, the factor reductions and the range stabilities move into `LocalCovField/`, stated for `ofFactors`

In order:

1. **Done.** `LocalGaugeData/` imports nothing above it (PR #1415 and the `LocalGaugeData/` PRs).
2. **Field data.** Rework PR #2011 as in §5, add `Field T`, settle conjugates and compatibility, then realize the symbols in $J(T)$.

3. **Sectors and the card.** Derive the sectors from maps of index types, and migrate the Standard Model from the hand-built field data to the card.

4. **After the deadline.** Replace the real connection tower by the field data with no matter, last, because the Standard Model gauge-sector classification is built on it; add reality and a total-derivative derivation commuting with the gauge action, so that invariants can be classified modulo total derivatives.

Steps 2 and 3 are the target for early December.

7 Progress

`LocalGaugeData/` (September 2026, PR #1415)

The layer imported matter in two ways; both are gone. The factor types `U1Factor`, `SUFactor`, `Factor`, `Factors` moved down into `Factor.lean`, since their fields mention only gauge jets, the gauge algebra and the jet ring. `TransformsIn` and `InfinitesimalAction`, statements about a representation on `JetRing` $\otimes$ V, moved up to `MatterField/`. The duplication between `U1.lean` and `SU/Basic.lean` was removed by `MatrixJets.lean`, a presentation of gauge jets by matrices of jets from which the laws, faithfulness and the canonical `SU`-type factor are proved once. `MatrixJets.free` (Joseph, 25 September) reduces `Free` to three conditions on the carriers, and the Standard Model freeness proof became an instance, about 400 lines shorter. The layer is eleven files, about 3 600 lines, importing nothing outside itself: `Basic`, `AdjointCoeff`, `MaurerCartan`, `Truncation`, `Factor`, `MatrixJets`, `SU/Algebra`, `SU/Basic`, `U1`, `Prod`, `OfFactors`.

Not done: `U1.lean` and `SU/Basic.lean` still define their structure maps by hand. Defining the carriers as subtypes of matrices would change the types the Standard Model files unfold to, so it waits for the hand-built Standard Model code to go.

Matter fields and species (PRs #1989 and #2011, October–November 2026)

PR #1989 (merged 30 October) added `statistic : FieldStatistic` to `MatterField` and requires the fermionic and bosonic entries of the field data to match; merging fields now needs a common statistic as well as a common mass weight. PR #2011 (opened 1 November) introduced a species type with case-defined coefficients and gauge, weight, derivative and Lorentz laws, and moved the coefficients into `MatterField/RepCoeff` and `GaugeBoson/GaugeCoeff`. After discussion on 2 November the species type is dropped and the PR is to be reworked as in §5.